# Cryptocurrency Pairs Trading Research

Cointegration screening, rolling z-score signals and historical backtesting using Binance market data

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Public academic report extract

This public version removes university administrative pages and personal candidate identifiers. The underlying assignment also includes related momentum, portfolio optimisation and fixed-income analysis that provide context for the cryptocurrency strategy research.

# Introduction

This report evaluates the integration of two algorithmic crypto trading strategies—momentum and statistical arbitrage—into the University of Exeter's (UE) high-risk investment portfolio. It combines bond yield modelling, time series analysis, and portfolio optimisation, leveraging financial theory and empirical data to assess performance, manage risk, and identify viable improvements.

# Data

**Bond Portfolio:**

| Bond | B1 | B2 | B3 | B4 |
|---|---|---|---|---|
| Maturity (years) | 2 | 8 | 17 | 30 |
| Coupon | 3.00% | 4.00% | 3.50% | 2.50% |
| Weight | 25% | 25% | 25% | 25% |

Figure 1: UE's Current Bond Portfolio

As Central banks have begun to lower interest rates it is important to analyse UE's current bond position and potential impacts caused by these changes. 27 other AAA-rated corporate bonds with maturities between 1 and 20 years, coupon rates ranging from 2.50% to 6.25%, were used to determine discount factors and then create a 20 year spot yield curve[1]:

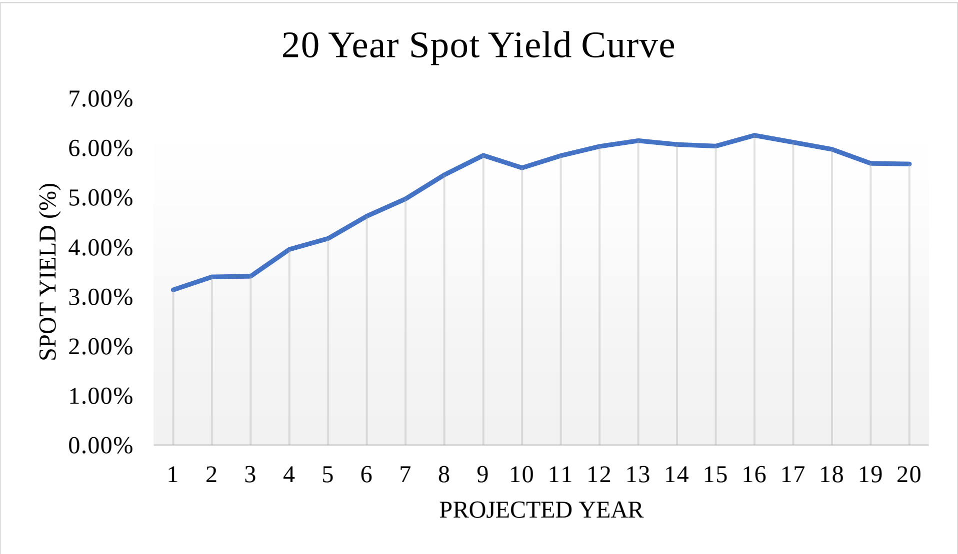


Figure 2: 20 Year Spot Yield Curve

The yield spread between 10-year and 1-year maturities is 2.46%, reflecting expectations of rising interest rates (Per, 2018).

[1] See Appendix A for Table of Spot Yields

**Cryptocurrency Strategy Data:**

Daily closing prices for Bitcoin (BTC) and Ethereum (ETH), covering the period from 01/01/2018 to 31/12/2024, were collected and converted into daily returns for further analysis. Summary statistics derived from these returns offer clear insight into the performance characteristics of these potential portfolio assets

BTC Annualised Daily Returns Summary Statistics

| Year | Mean | Standard Deviation | Sharpe Ratio | Skewness | Kurtosis |
|---|---|---|---|---|---|
| 2018 | -0.2633% | 0.0424 | -0.0621 | -0.2136 | 1.7945 |
| 2019 | 0.2417% | 0.0356 | 0.0679 | 0.5620 | 4.6981 |
| 2020 | 0.4582% | 0.0377 | 0.1215 | -2.2061 | 28.3717 |
| 2021 | 0.2165% | 0.0421 | 0.0514 | 0.1132 | 1.5608 |
| 2022 | -0.2255% | 0.0333 | -0.0677 | -0.2912 | 4.3004 |
| 2023 | 0.2830% | 0.0229 | 0.1236 | 0.8472 | 3.3117 |
| 2024 | 0.2558% | 0.0280 | 0.0912 | 0.4898 | 1.9598 |

Figure 3: BTC Annualised Daily Returns Summary Statistics

ETH Annualised Daily Returns Summary Statistics

| Year | Mean | Standard Deviation | Sharpe Ratio | Skewness | Kurtosis |
|---|---|---|---|---|---|
| 2018 | -0.3221% | 0.0560 | -0.0575 | -0.0250 | 1.1761 |
| 2019 | 0.0771% | 0.0411 | 0.0188 | -0.1208 | 3.2479 |
| 2020 | 0.6070% | 0.0494 | 0.1229 | -1.5065 | 16.5396 |
| 2021 | 0.5983% | 0.0560 | 0.1068 | 0.0089 | 3.4818 |
| 2022 | -0.2043% | 0.0453 | -0.0451 | -0.0529 | 2.7340 |
| 2023 | 0.2065% | 0.0244 | 0.0845 | 0.5158 | 2.4588 |
| 2024 | 0.1606% | 0.0341 | 0.0471 | 0.7550 | 4.1092 |

Autocorrelation (ACF) and partial autocorrelation (PACF) functions for BTC and ETH daily returns (2018-2022) indicate evidence of serial correlation. BTC exhibits significant autocorrelation primarily at lag 1, whereas ETH returns show fluctuating but statistically significant autocorrelations at multiple early lags (up to lag 10). These patterns suggest the potential suitability of low-order ARIMA models for subsequent modelling.

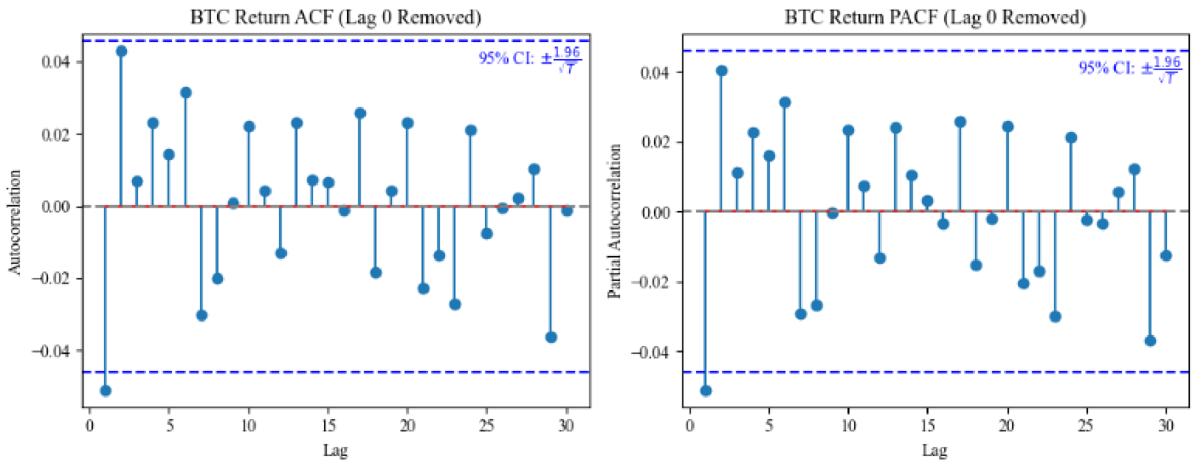


Figure 5: BTC Autocorrelation & Partial Autocorrelation Plots

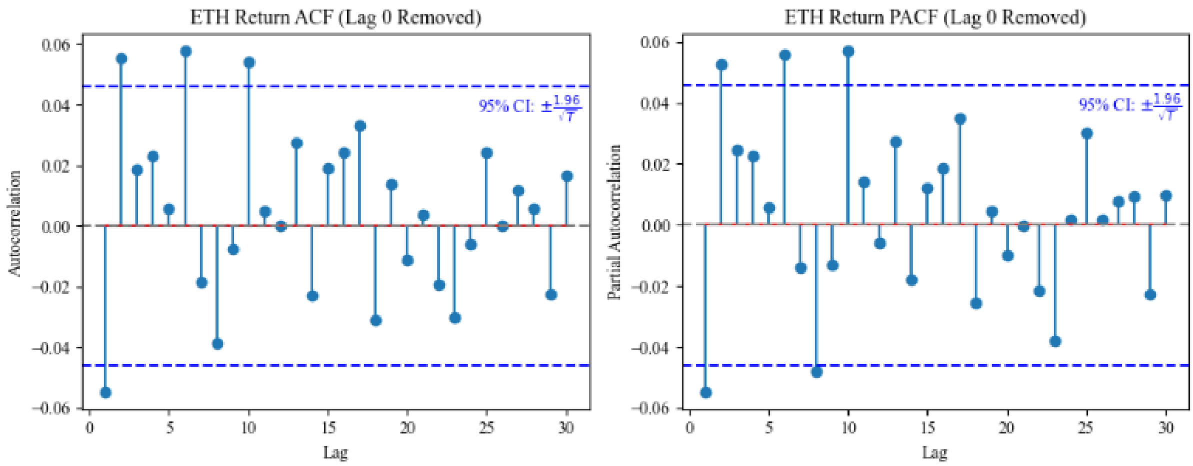


Figure 6: ETH Autocorrelation & Partial Autocorrelation Plots

A Dickey-Fuller test assessed the suitability of Bitcoin and Ethereum as trading pairs for statistical arbitrage by testing cointegration on the price spread (Dickey & Fuller, 1979). The resulting test statistic for the spread was -2.2165, above the 5% critical value threshold of -2.86 (Bhandari et al., 2017), indicating insufficient evidence of cointegration at the 5% significance level[2].

**High-Risk Portfolio Composition:**

UE's high-risk portfolio comprises a selection of ETFs, with asset weights optimised to maximise the portfolio's Sharpe Ratio (SR)—a measure of risk-adjusted return (Sharpe, 1994)—without imposing short-selling constraints.

| Portfolio Zero ($P_0$): | VGK | VOO | USO | GLD | IYR | GBTC |
|---|---|---|---|---|---|---|
| Portfolio Weights | -121.69% | 197.89% | 20.19% | 77.18% | -78.08% | 4.52% |
| Average Monthly Return | 0.72% | 1.19% | 1.13% | 0.79% | 0.65% | 4.32% |
| Average Monthly St. Dev | 0.0613 | 0.0561 | 0.1474 | 0.0419 | 0.0599 | 0.2463 |
| Average Monthly SR | 0.1172 | 0.2118 | 0.0764 | 0.1877 | 0.1093 | 0.1755 |

Figure 7: High Risk Portfolio Zero ($P_0$)

$P_0$ Summary Statistics:

| | |
|---|---|
| Monthly Expected Return | 1.995% |
| Port Variance | 0.0025 |
| Port St. Dev. | 0.0504 |
| Port Sharpe Ratio | 0.3962 |
| Total Weight | 100% |

Figure 8: Portfolio Zero ($P_0$) Summary Statistics

[2] See Appendix B for full Dickey-Fuller Test Results

# Methodology:

**Bond Portfolio Optimisation:**

To construct a smooth and interpretable yield curve for pricing and risk assessment, the Nelson-Siegel (NS) model was fitted to the calculated log spot yields. This functional form mitigates market noise often present in raw yield curves and captures short-, medium-, and long-term dynamics through four parameters: $\alpha_1$, $\alpha_2$, $\alpha_3$, and the decay rate β (Benninga and Tal Mofkadi, 2022). Optimal parameters were estimated by minimising the sum of squared errors between the log of observed and fitted yields across maturities.

$$y(t) = \alpha_1 + \alpha_2 \left( \beta \, \frac{1 - e^{-t/\beta}}{t} \right) + \alpha_3 \left( \beta \, \frac{1 - e^{-t/\beta}}{t} - e^{-t/\beta} \right)$$

Equation 1: Nelson-Siegel Model Formula

Using the fitted NS curve, a Monte Carlo simulation was conducted to model a potential interest rate cut scenario (Metropolis & Ulam, 1949), adjusting the log yield via a stochastic shock as defined in Equation 2. Simulated yield paths were used to reprice each bond (B1–B4) and compute their expected returns, allowing for evaluation of the interest rate shock's impact on UE's bond portfolio.

$$y_t = f(t) - D \times [0.0026 - 0.00015 \ln(1 + t)] + e_t$$

Where $D \sim Bernoulli$ (0.75), $e_t \sim N\left(0, \widehat{\sigma^2}\right)$ from Equation 1.

Equation 2: Interest Rate Cut via Stochastic Shock

Finally, based on the simulated returns, an optimal bond portfolio was constructed by estimating expected returns and the variance-covariance matrix. Portfolio weights were optimised to maximise the Sharpe Ratio, assuming no short selling.

**Momentum Strategy:**

An ARIMA model was fitted to BTC daily returns from 2018–2022 to capture time series dependencies. Model selection was based on minimising the Akaike Information Criterion (AIC), with candidate specifications limited to ARIMA(p,0,q) where p,q ∈[1,3]. The optimal model was identified as ARIMA(1,0,1), with an AIC of -6716.96, indicating a good fit (Leitner & Turner, 2016).

The strategy generates signals using the most recent 60 return observations to estimate an AR(2) model, producing a forecasted return $\hat{r}^t$. If $\hat{r}^t$ > max(0,$(\bar{r})_{360}$), where $(\bar{r})_{360}$ is the average return over the past 360 days, the strategy initiates a buy order; otherwise, no position is taken. Daily cumulative returns were aggregated to monthly intervals from 2019–2024 for performance evaluation.

**Statistical Arbitrage Strategy:**

To assess the viability of BTC and ETH as a trading pair, the price spread was defined (Equation 3), with the hedge ratio $\alpha$ approximated using the average ratio of BTC to ETH closing prices from 2018 to 2022. This yielded a static coefficient of 13.0637.

$$\text{Spread}_t = \text{BTC}_t - \alpha \cdot \text{ETH}_t$$

Equation 3: Spread Construction for Statistical Arbitrage Strategy

The arbitrage strategy follows a set simple z-score-based threshold rules:

If $z_t < -1$: Buy BTC and Short α ETH (Eq. 4a)

If $z_t > -1$: Short BTC and Buy α ETH (Eq. 4b)

If $|z_t| \leq 1$: Close all positions (Eq. 4c)

Equation 4: Arbitrage Strategy Threshold Rules

To assess potential implementation of these new high-risk strategies into UE's high-risk portfolio a new optimal portfolio (P1) is constructed by including the two trading strategies alongside the current ETFs in P0.

# Results:

**Bond Portfolio Results:**

To mitigate noise in the raw spot yield curve, NS model was fitted to the log yields. The fitted curve closely tracks the observed data and formed the optimal parameters below:

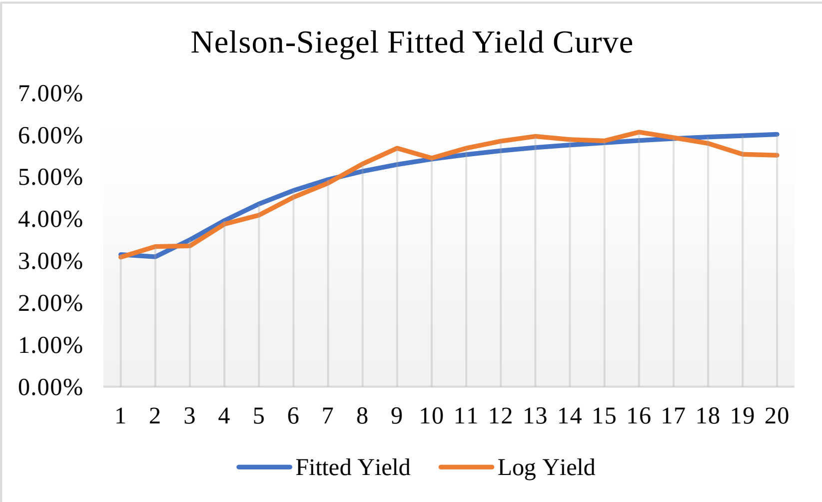


Figure 9: Nelson-Siegel Fitted Yield Curve

Fitted Yield Curve Optimal Parameters

| | |
|---|---|
| $\alpha_1$ | 0.066034 |
| $\alpha_2$ | -0.017641 |
| $\alpha_3$ | -0.090449 |
| β | 1.094486 |

Figure 10: Fitted Yield Curve Optimal Parameters

These parameters reflect a steep initial yield level with moderate medium- and long-term curvature, supporting the model's suitability for extrapolating yields up to 30 years.

A Monte Carlo simulation incorporating the assumed rate cut was used to reprice bonds (1-4) and compute expected returns. Across 1,000 simulations, the average portfolio return sat at 1.73%, with bond B4 showing the largest return due to its duration sensitivity. The simulated yield curve shifted downwards, reflecting the anticipated decline in rates.

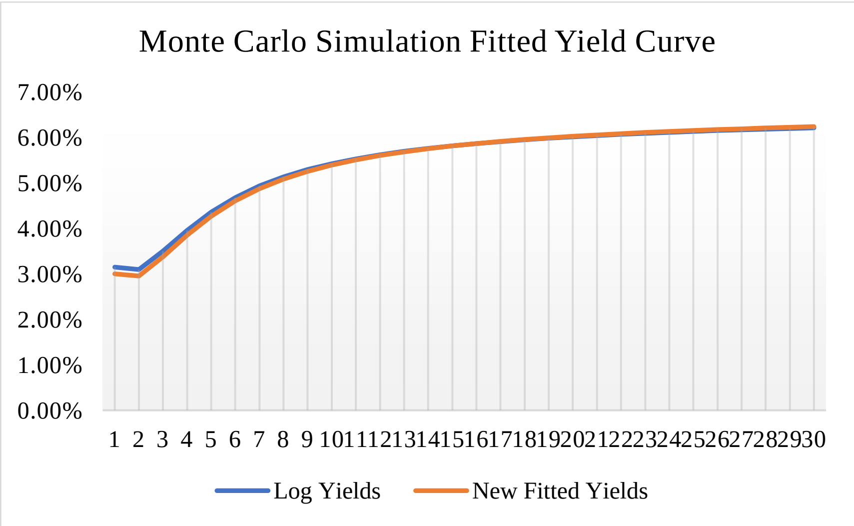


Figure 11: Monte Carlo Simulation Fitted Yield Curve

| Original Bond Portfolio (P0) Summary Statistics: | | B1 | B2 | B3 | B4 | Portfolio Average |
|---|---|---|---|---|---|---|
| Weights | | 25% | 25% | 25% | 25% | 25% |
| NS (Eq.1) | FP: | 99.73 | 92.71 | 74.59 | 50.63 | 79.41 |
| | YTM: | 3.14% | 5.13% | 5.87% | 6.14% | 5.07% |
| MC Sim (Eq.2) | Mean FP: | 100.09 | 93.83 | 76.50 | 52.04 | 80.61 |
| | Mean YTM: | 2.95% | 4.95% | 5.69% | 5.98% | 4.89% |
| Mean Return (MC Sim) | | 0.37% | 1.21% | 2.56% | 2.79% | **1.73%** |

Figure 12: Summary Statistics – Original Bond Portfolio (P0)

Based on the simulated returns, an optimal portfolio was then constructed with nearly 100% of capital being allocated to bond B3, which offered the most attractive risk-adjusted return (SR = 2.2343). This allocation aligns with expectations in a declining rate environment, favouring longer-duration bonds.

| Optimal MC Bond Portfolio (P2) Summary Statistics: | B1 | B2 | B3 | B4 | Portfolio |
|---|---|---|---|---|---|
| Optimal Portfolio Weights | 0.39% | 0.00% | 99.53% | 0.08% | 100.00% |
| Average St. Dev | 0.0020 | 0.0067 | 0.0115 | 0.0156 | 0.0115 |
| Average Sharpe Ratio | 1.8080 | 1.8079 | 2.2343 | 1.8081 | 2.2335 |
| Average Return | 0.37% | 1.22% | 2.58% | 2.81% | **2.57%** |

Figure 13: Summary Statistics – Optimised Bond Portfolio (P2)

The optimised portfolio produced an average return of 2.57%, up from 1.73% in the equally weighted case, with an excellent Sharpe ratio, demonstrating the value of reallocation under the simulated yield environment.

**Crypto Strategy Performance:**

Momentum Strategy:

Average monthly returns show the strategy delivered consistent positive returns in most years, with the exception of 2022, where it underperformed due to rapidly changing market direction. The annualised Sharpe ratios ranged from 0.36 to 0.70, reflecting modest risk-adjusted performance.

| Cumulative Monthly Momentum Returns | | | |
|---|---|---|---|
| Year | Mean | S.D. | Sharpe |
| 2019 | 7.89% | 0.1603 | 0.4920 |
| 2020 | 8.50% | 0.1218 | 0.6979 |
| 2021 | 6.48% | 0.1083 | 0.5983 |
| 2022 | -4.91% | 0.1291 | -0.3805 |
| 2023 | 3.74% | 0.0610 | 0.6136 |
| 2024 | 3.79% | 0.1047 | 0.3618 |
| 2019~24 | 4.25% | 0.1224 | 0.3469 |

Figure 14: Summary Statistics – Momentum Strategy Annualised Monthly Returns

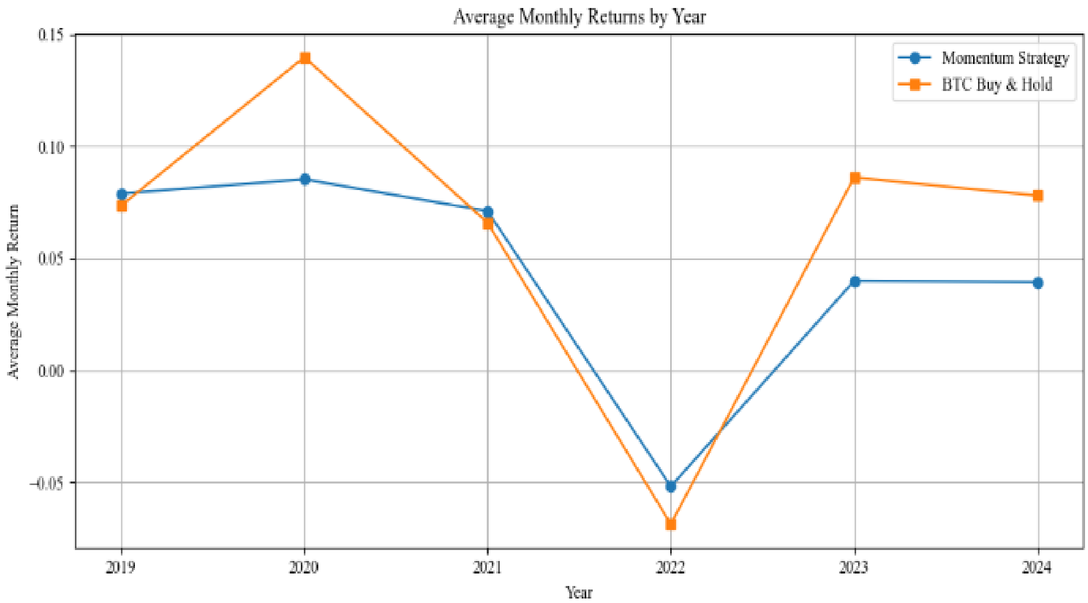


Figure 15: Average Monthly Returns by Year – Momentum vs BTC Buy-and-Hold

While the Buy-and-Hold BTC strategy delivered higher returns overall, it exhibited significantly greater volatility and deeper drawdowns. In contrast, the momentum strategy's smoother return profile and stronger Sharpe ratios in several years support its role as a stabilising component within UE's high-risk portfolio.

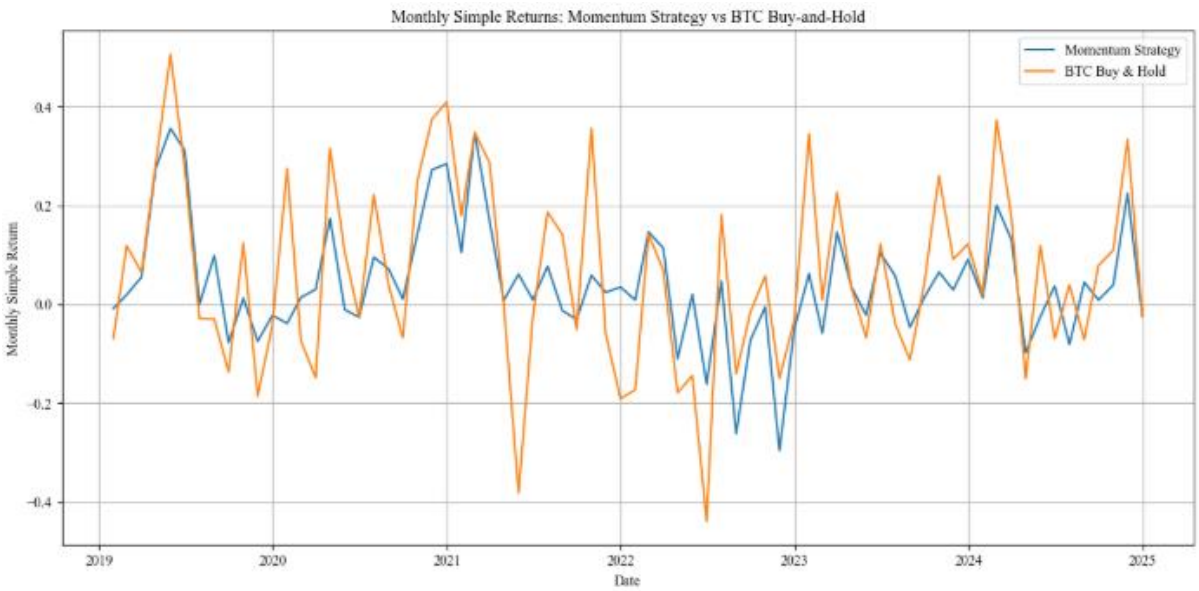


Figure 16: Monthly Returns – Momentum vs BTC Buy-and-Hold

The compounded cumulative returns plot further illustrates the momentum strategy's lower volatility, with a steadier growth path compared to the buy-and-hold strategy, which exhibits more frequent and severe fluctuations over the same period.

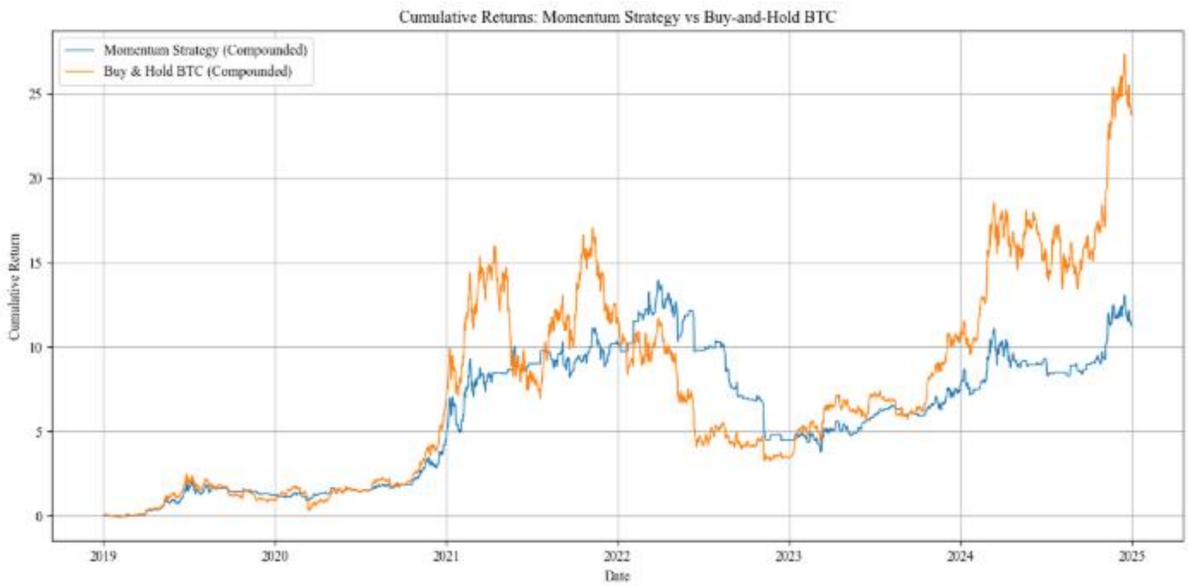


Figure 17: Compounded Cumulative Returns – Momentum vs BTC Buy-and-Hold

Statistical Arbitrage Strategy:

Monthly return performance over 2019–2024 shows high volatility and inconsistent returns. The strategy exhibits extreme spikes and drawdowns, particularly between 2022 and 2024, indicating instability in spread convergence.

Statistical Arbitrage Summary Statistics

| Year | Mean | S.D. | Sharpe |
|---|---|---|---|
| 2019 | 0.00% | 0 | N/A |
| 2020 | 0.03% | 0.0024 | 0.1244 |
| 2021 | 19.37% | 0.6476 | 0.2992 |
| 2022 | -48.00% | 1.9523 | -0.2459 |
| 2023 | 13.26% | 1.2932 | 0.1025 |
| 2024 | -8.23% | 0.1682 | -0.4894 |
| 2019~24 | -3.93% | 0.9833 | -0.0400 |

Figure 18: Summary Statistics – Statistical Arbitrage Strategy

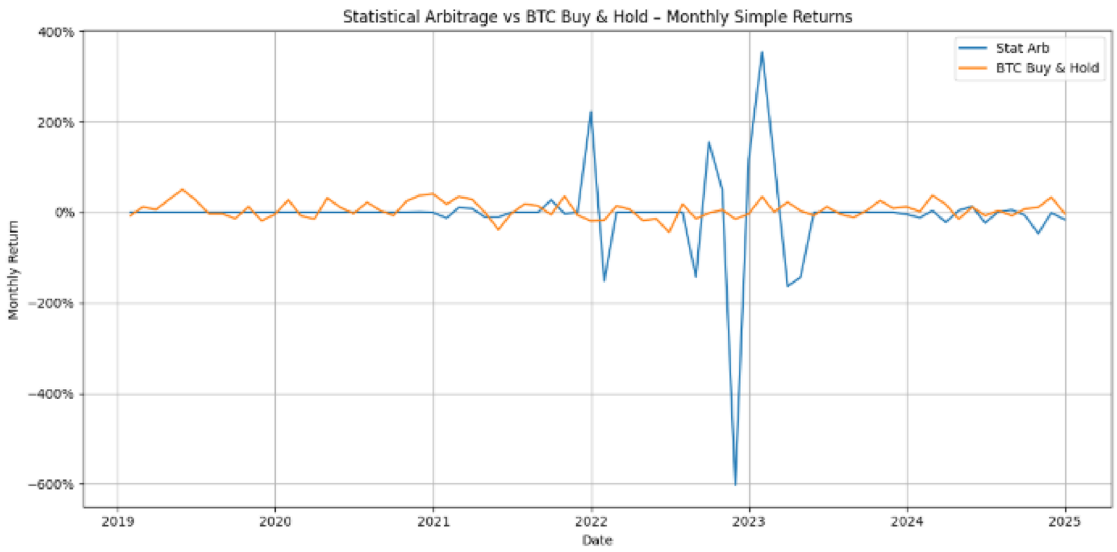


Figure 19: Monthly Returns – Stat Arb vs BTC Buy-and-Hold

While the strategy generated positive returns in selected years, it suffered heavy losses in 2022 and 2024, leading to a negative cumulative return over the full period. The low and negative Sharpe ratios, combined with significant volatility and instability, suggest that the strategy as currently implemented is not a reliable contributor to portfolio performance.

**Strategy Comparison:**

The momentum strategy shows stronger consistency and risk-adjusted performance from 2019–2024, with a Sharpe ratio of 0.35 and only one year of negative returns. In contrast, the statistical arbitrage strategy delivered a –3.93% cumulative return and Sharpe ratio of –0.04, indicating poor stability. Figure 20 highlights this disparity, as momentum and buy-and-hold strategies remain relatively stable, while the stat arb strategy displays extreme spikes and collapses, signalling failure of spread convergence.

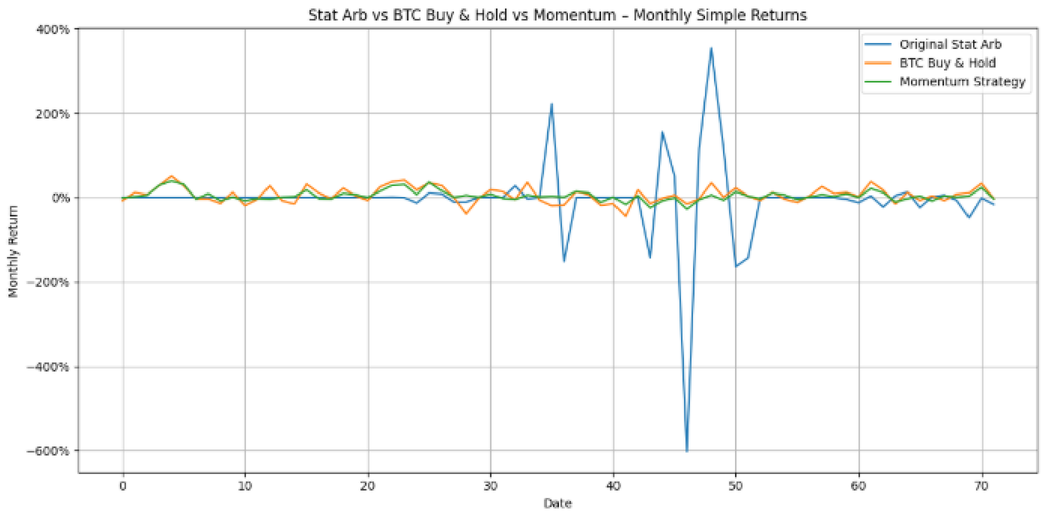


Figure 20: Monthly Returns – Stat Arb vs BTC Buy-and-Hold

The instability of the statistical arbitrage results can largely be attributed to the absence of cointegration between BTC and ETH. Without a mean-reverting spread, the assumption that price differences will normalise over time breaks down, leading to substantial losses when positions fail to close profitably. This structural weakness suggests the strategy, as implemented, is unlikely to be robust without further refinement.

**Portfolio Integration Analysis:**

P0 achieved a monthly return of 1.995% and a Sharpe ratio of 0.396. However, it relied on highly leveraged positions, including –121.7% in VGK and 197.9% in VOO, raising concerns regarding its real-world viability.

To assess the value of integrating the prospective trading strategies, a new portfolio (P1) was optimised over the same 2019–2022 period. The optimiser allocated 46.5% to momentum, while statistical arbitrage was effectively excluded (–3.3%) due to its negative Sharpe ratio and poor performance. Traditional asset exposures were rebalanced, with reductions in VGK, IYR, and GBTC.

| Portfolio 1 ($P_1$): | Portfolio Weights | Average Monthly Return | Average Monthly St. Dev | Average Monthly Sharpe Ratio |
|---|---|---|---|---|
| VGK | -167.39% | 0.72% | 0.0613 | 0.1172 |
| VOO | 215.61% | 1.19% | 0.0561 | 0.2118 |
| USO | 21.14% | 1.13% | 0.1474 | 0.0764 |
| GLD | 55.20% | 0.79% | 0.0419 | 0.1877 |
| IYR | -53.53% | 0.65% | 0.0599 | 0.1093 |
| GBTC | -14.24% | 4.32% | 0.2463 | 0.1755 |
| Momentum | 46.53% | 4.49% | 0.1385 | 0.3240 |
| Stat Arb | -3.31% | -7.15% | 1.0264 | -0.0697 |

Figure 21: High Risk Portfolio One ($P_1$)

$P_1$ Summary Statistics:

| | |
|---|---|
| Monthly Expected Return | 3.39% |
| Port Variance | 0.0038 |
| Port St. Dev. | 0.0617 |
| Port Sharpe Ratio | 0.5492 |
| Total Weight | 100% |

Figure 22: Portfolio Zero ($P_1$) Summary Statistics

P1 outperformed P0 across all metrics, delivering a higher monthly return (3.39%), slightly increased volatility (6.17%), and an improved Sharpe ratio of 0.549. The shift in allocations is economically intuitive: the large momentum allocation reflects its consistent, low-volatility returns, making it a valuable diversifier. The exclusion of statistical arbitrage aligns with its poor Sharpe ratio (–0.04) and erratic behaviour. Continued short positions in VGK, IYR, and GBTC reflect their weak risk-adjusted performance or high correlation with underperforming assets.

To assess implementability under realistic leverage limits, a third portfolio (P2) was constructed by applying a gross exposure constraint ($\sum|\omega\_i| = 1$). While leverage was significantly reduced, the portfolio still achieved a 1.70% return and a Sharpe ratio of 0.417. Though lower than P1, it outperformed P0, reinforcing that momentum remains a valuable addition—even under realistic investment constraints.

| Portfolio Adjustments: | Portfolio 0 Weights | Portfolio 1 Weights | Portfolio 2 Weights |
|---|---|---|---|
| VGK | -121.69% | -167.39% | -14.75% |
| VOO | 197.89% | 215.61% | 0.00% |
| USO | 20.19% | 21.14% | 15.38% |
| GLD | 77.18% | 55.20% | 16.28% |
| IYR | -78.08% | -53.53% | -11.81% |
| GBTC | 4.52% | -14.24% | -10.20% |
| Momentum | N/A | 46.53% | 28.44% |
| Stat Arb | N/A | -3.31% | -3.15% |
| Absolute Investment | 499.55% | 576.95% | 100.00% |

Figure 23: Asset Weight Comparison – P0 vs P1 vs P2

| Portfolio Comparison: | P0 | P1 | P2 |
| --- | --- | --- | --- |
| Monthly Expected Return | 1.995% | 3.390% | 1.699% |
| Port Variance | 0.0025 | 0.0038 | 0.0017 |
| Port St. Dev. | 0.0504 | 0.0617 | 0.0407 |
| Port Sharpe Ratio | 0.3962 | 0.5492 | 0.4169 |

Figure 24: Portfolio Comparison– P0 vs P1 vs P2

## Improvements:

The original BTC/ETH statistical arbitrage strategy failed the cointegration test, weakening its theoretical justification. To improve robustness, a revised framework was implemented based on EdgeTrader (2020), incorporating updated crypto data, systematic cointegration testing, and refined strategy logic[3].

The new framework identified BTC/XRP as a viable pair (p = 0.0146), and a revised stat-arb strategy was constructed using this relationship. Several versions were explored, including a hybrid model with a rolling 60-day mean. The optimised version (BTC/XRP) delivered significantly improved total returns (43.1% vs –3.9%) and a marginally positive Sharpe ratio (0.018 vs –0.040).

Original Stat Arb - ETH

| Year | Mean | S.D. | Sharpe |
|---|---|---|---|
| 2019 | 0.00% | 0 | N/A |
| 2020 | 0.03% | 0.0024 | 0.1244 |
| 2021 | 19.37% | 0.6476 | 0.2992 |
| 2022 | -48.00% | 1.9523 | -0.2459 |
| 2023 | 13.26% | 1.2932 | 0.1025 |
| 2024 | -8.23% | 0.1682 | -0.4894 |
| 2019~24 | -3.93% | 0.9833 | -0.0400 |

Figure 25: Original Stat Arb Summary Statistics - ETH

New Stat Arb - XRP

| Year | Mean | S.D. | Sharpe |
|---|---|---|---|
| 2019 | -398.18% | 10.564745 | -0.376895 |
| 2020 | 705.35% | 14.9995 | 0.4702 |
| 2021 | 717.52% | 15.4643 | 0.4640 |
| 2022 | 248.27% | 10.4080 | 0.2385 |
| 2023 | -916.02% | 50.0062 | -0.1832 |
| 2024 | -98.08% | 15.5009 | -0.0633 |
| 2019~24 | 43.14% | 23.7827 | 0.0181 |

Figure 26: New Stat Arb Summary Statistics - XRP

However, the strategy continues to exhibit extreme volatility, with several large drawdowns. This suggests that while the strategy has high upside potential, further refinements—such as implementing a stop-loss mechanism—could help mitigate severe losses and stabilise returns.

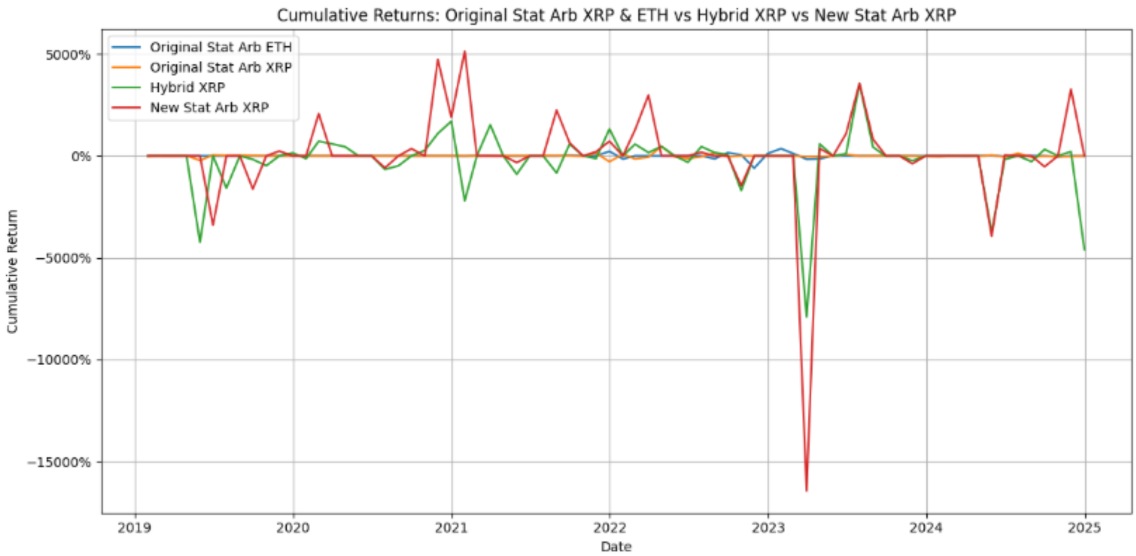


Figure 27: Performance Comparison of Original and Improved Stat Arb Models

Integrating the improved strategy into the high-risk portfolio led to meaningful performance enhancements. Portfolio 3 (P3) saw Sharpe improve from 0.396 (P0) to 0.585. A similarly constrained version (P4), still delivered a Sharpe of 0.452—demonstrating that most of the benefit can be retained without excessive leverage.

| Improved Optimised Portfolios | Portfolio 3 Weights | Portfolio 4 Weights | Average Monthly Return | Average Monthly St. Dev | Average Monthly Sharpe Ratio |
|---|---|---|---|---|---|
| VGK | -100.84% | 2.82% | 0.72% | 0.0613 | 0.1172 |
| VOO | 170.45% | 0.01% | 1.19% | 0.0561 | 0.2118 |
| USO | 11.96% | 3.86% | 1.13% | 0.1474 | 0.0764 |
| GLD | 69.15% | 54.07% | 0.79% | 0.0419 | 0.1877 |
| IYR | -65.34% | -7.47% | 0.65% | 0.0599 | 0.1093 |
| GBTC | -9.44% | -5.75% | 4.32% | 0.2463 | 0.1755 |
| Momentum | 23.84% | 25.84% | 4.49% | 0.1385 | 0.3240 |
| New Stat Arb | 0.22% | 0.18% | 318.24% | 13.4620 | 0.2364 |

Figure 28: High Risk Portfolio 3 & 4 Weights

| Portfolio Comparison: | P0 | P1 | P2 | P3 | P4 |
|---|---|---|---|---|---|
| Monthly Expected Return | 1.995% | 3.390% | 1.699% | 2.911% | 1.918% |
| Port Variance | 0.0025 | 0.0038 | 0.0017 | 0.0025 | 0.0018 |
| Port St. Dev. | 0.0504 | 0.0617 | 0.0407 | 0.0498 | 0.0424 |
| Port Sharpe Ratio | 0.3962 | 0.5492 | 0.4169 | 0.5851 | 0.4518 |

Figure 29: High Risk Portfolio Comparison Across Strategies

## Conclusion:

The momentum strategy proved a valuable portfolio addition, enhancing risk-adjusted returns. However, the original stat-arb model lacked validity due to non-cointegration. An improved XRP-based strategy showed promise but remained volatile. Overall, portfolio performance improved with refined strategy inclusion, though further risk controls are recommended for future use.

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# Appendix:

**Appendix A: Spot Yield Table**

| Year | Spot Yields |
|---|---|
| 1 | 3.13% |
| 2 | 3.39% |
| 3 | 3.41% |
| 4 | 3.95% |
| 5 | 4.17% |
| 6 | 4.62% |
| 7 | 4.97% |
| 8 | 5.45% |
| 9 | 5.85% |
| 10 | 5.60% |
| 11 | 5.84% |
| 12 | 6.03% |
| 13 | 6.14% |
| 14 | 6.07% |
| 15 | 6.03% |
| 16 | 6.25% |
| 17 | 6.11% |
| 18 | 5.97% |
| 19 | 5.69% |
| 20 | 5.67% |

Figure 30: Computed Spot Rates (Derived from Discounting Cash Flows)

**Appendix B: Dickey-Fuller Test Results**

Dicky-Fuller Test

| DF Test | | BTC | ETH |
|---|---|---|---|
| Coefficients | | | |
| | Lag y | -0.0025 | -0.0036 |
| | Time | 0.0333 | 0.0048 |
| | Constant | 21.3418 | 0.0214 |
| # of obs. | | 1825 | 1825 |
| Resid. Variance | | 1016476.32 | 6381.04 |
| Coef. Variance | | | |
| | Lag y | 0.0000 | 0.0000 |
| | Time | 0.0035 | 0.0000 |
| | Constant | 2233.2713 | 14.1782 |
| DF-stat | | | |
| | Lag y | -1.3403 | -1.7537 |
| | Time | 0.5616 | 1.0166 |
| | Constant | 0.4516 | 0.0057 |

Figure 31: Full Dickey-Fuller Test Result

Dicky-Fuller Test

| | | Spread |
|---|---|---|
| Coefficients | | |
| | Lag y | -0.0056 |
| | Time | -0.0012 |
| | Constant | 29.7153 |
| # of obs. | | 1825 |
| Resid. Variance | | 488106.1457 |
| Coef. Variance | | |
| | Lag y | 0.0000 |
| | Time | 0.0010 |
| | Constant | 1159.7813 |
| DF-stat | | |
| | Lag y | -2.2165 |
| | Time | -0.0396 |
| | Constant | 0.8726 |

Figure 32: Full Dickey-Fuller Test Result on Spread

**Appendix C: Stat Arb Strategy Improvements**

Statistical Arbitrage Strategy Improvements:

This appendix details the enhancements made to the original statistical arbitrage (stat arb) strategy, incorporating a new asset pair, improved signal logic, and parameter optimisation, all implemented in Python using a custom back testing engine.

1. Motivation for Improvement

The initial BTC/ETH strategy performed poorly over the 2019–2024 period, delivering a negative Sharpe ratio (–0.04) and failing the Augmented Dickey-Fuller (ADF) cointegration test. This invalidated the key assumption of mean-reverting price dynamics and justified a comprehensive model overhaul.

2. Data and Pair Selection

Using the Binance API (2025), daily close prices were collected for over 75 cryptocurrency pairs. After filtering for liquidity and correlation, each pair underwent ADF testing. BTC/XRP was identified as the most viable alternative, with a cointegration p-value of 0.0146. This pair had sufficient historical data and showed a stable long-term relationship.

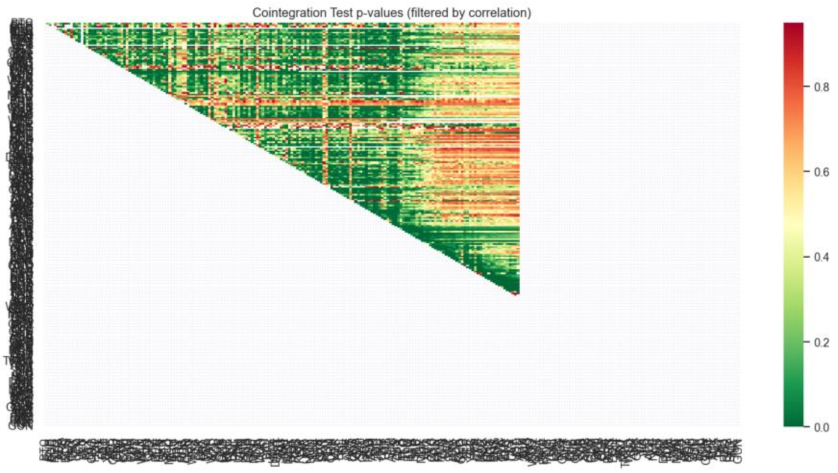



Figure 33: Cointegration Test P-Values (Filtered by Correlation)

3. Trading Model and Signal Logic

Building on the EdgeTrader (2020) GitHub framework, the spread was calculated as:

$$\text{Spread}_t = \log(\text{BTC}_t) - \beta \cdot \log(\text{XRP}_t)$$

Equation 5: Spread Equation

Where β is the static hedge ratio obtained via OLS regression.

The z-score was then computed as:

- Mean (60MA) - 60 days of moving average of ratio
- Standard Deviation (60SD) - 60 days standard deviation of ratio
- Latest Ratio (5MA) - 5 days of moving average of ratio
- Z-Score (Z) - (5MA - 60MA) / 60SD

And then Optimised:

$$Z = \frac{\text{Near\_MA} - \text{Far\_MA}}{\text{Std}(\text{Far\_MA})}$$

Equation 6: Z-Score Equation

- Entry Signals:
  - Long BTC / Short XRP if $Zt < -1$
  - Short BTC / Long XRP if $Zt > 1$

- Exit Signal:
  - Close position when $|Zt| < 0.1$

This logic was vectorised and executed daily over the sample period.


Figure 34: Price Ratio BTC/XRP


Figure 35: Z-Score Ratio BTC/XRP


Figure 36: Rolling Averages (5-day, 60-day)

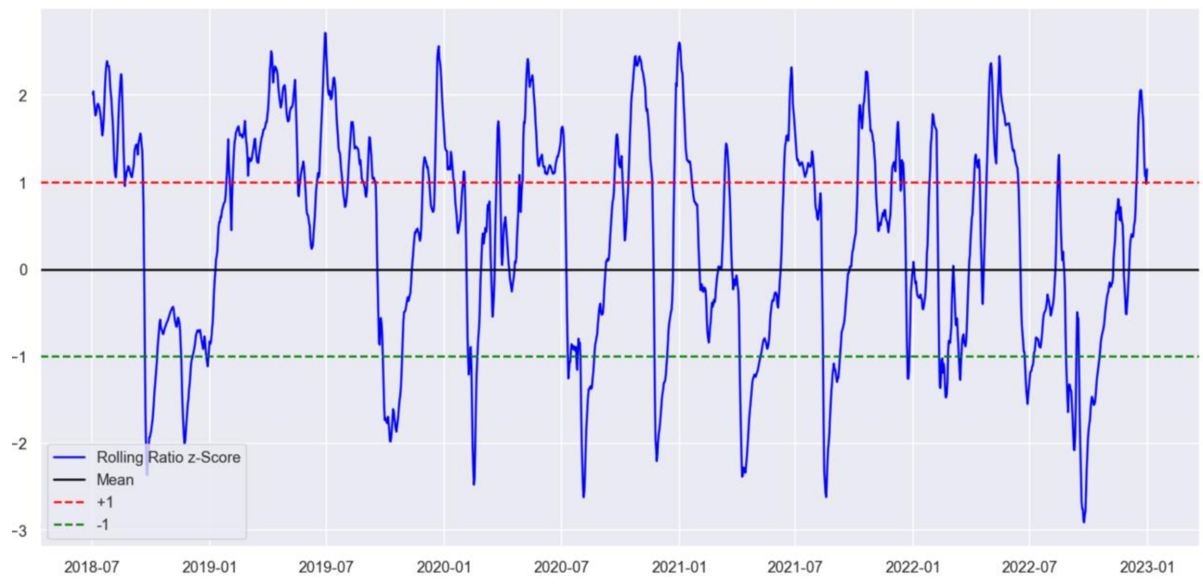



Figure 37: Rolling Ratio Z-Score


Figure 38: Transactions Overtime

4. Model Variants

Three versions were developed:

- Original: Rolling 60-day lookback for mean and standard deviation.
- Hybrid: Uses a static 60-day average with rolling volatility and z-scores.
- Optimised: Thresholds and window lengths were selected using grid search to maximise Sharpe ratio on the 2019–2022 training set.

5. Strategy Performance (2019–2024)

| Strategy | Total Return | Sharpe Ratio | Volatility (SD) |
|---|---|---|---|
| ETH (Original) | –3.93% | –0.040 | 0.983 |
| XRP (Original) | –3.06% | –0.037 | 0.829 |
| XRP (Hybrid) | –213.40% | –0.147 | 14.50 |

| Strategy | Total Return | Sharpe Ratio | Volatility (SD) |
|---|---|---|---|
| XRP (Optimised) | 43.14% | 0.018 | 23.78 |

Figure 39: Recap of Strategy Summary Statistics

The optimised strategy significantly outperformed others, albeit with elevated volatility and negative years (notably 2023). P&L plots and z-score mean reversion charts are provided in Figures X–Y.

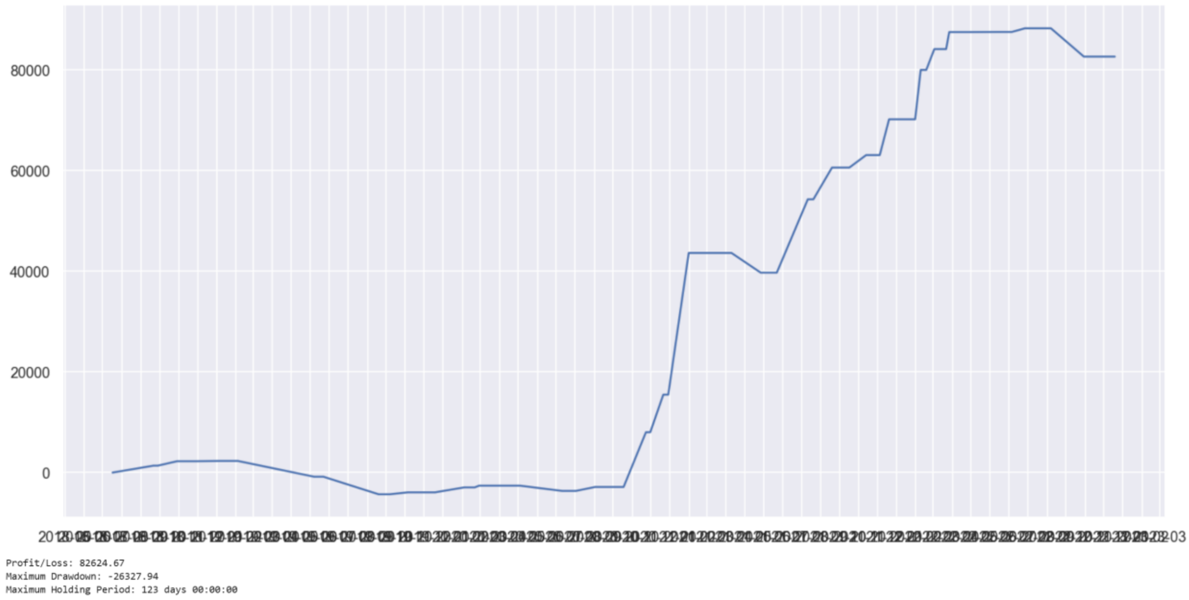
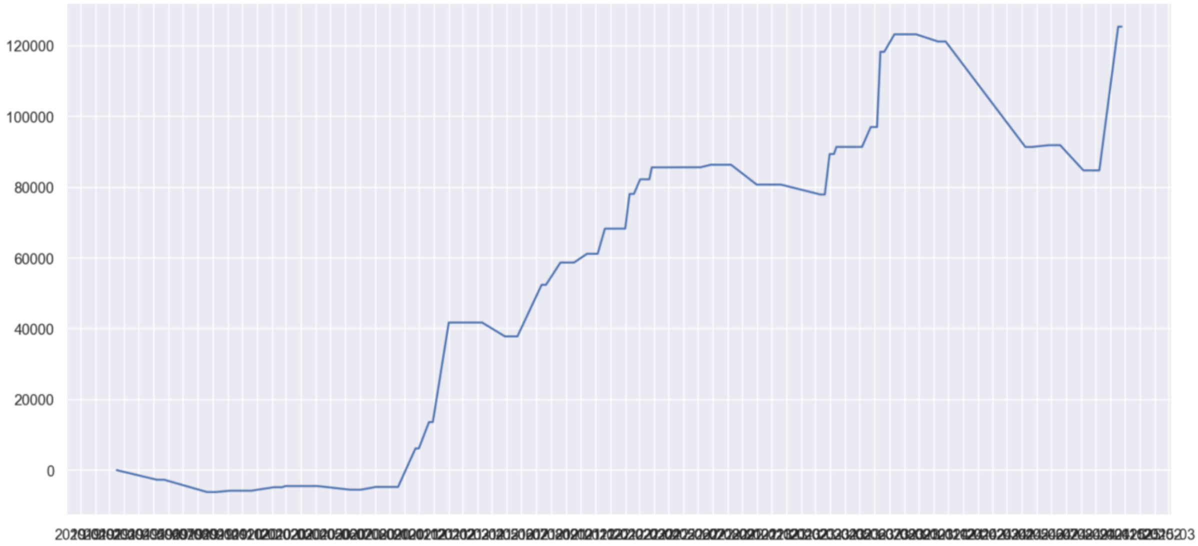



Figure 40: Trading on Training Data P&L (2018-2022)




```
Profit/Loss: 125375.50
Maximum Drawdown: -35130.35
Maximum Holding Period: 165 days 00:00:00
```

Figure 41: Out of Sample Trading Result P&L (2019-2024)

## 7. Recommendations for Further Development

Despite improved returns, the strategy remains volatile. Suggested future improvements include:

- Stop-loss filters to limit tail-risk drawdowns
- Dynamic position sizing based on z-score magnitude
- Volatility-adjusted signals to avoid overtrading in unstable periods